

Investigating Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

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Abstract

Methodology (The Pipeline)

Conclusion & Impact

- **Preprocessing Data:** Selected encounter-level identifiers, postoperative complication records, demographic variables, laboratory test results, and alignment timestamps (anesthesia start time and laboratory collection time) from the MOVER tables.
- **Cleaning Data:** Converted height and weight to metric units, imputed missing demographic values, and excluded implausible laboratory values. Laboratory predictors were selected as 5 prespecified tests and the 5 most common additional tests. For each target test, only the most recent preoperative result was retained for each encounter.
- **Feature Engineering:** Created a binary postoperative hypoxemia outcome from complication records, recoded sex as binary, and calculated BMI from height and weight.
- **Data Restructuring:** Renamed selected variables and reshaped laboratory data from long to wide format, creating one encounter-level row with separate columns for each laboratory predictor.
- **Data Validation:** The dataset was checked for duplicate keys, required columns, variable types, implausible values, and missingness. Variables with more than 50% missingness were dropped; pH, CRP, and lactate were excluded.
- **Cohort Split:** A case-control cohort was constructed by retaining all 98 hypoxemia encounters and randomly sampling 200 non-hypoxemia encounters, followed by a stratified 75%/25% training-test split.
- **Plots & Transformation:** Training-set exploratory plots were generated, and glucose and leukocytes were log-transformed.
- **Analysis:** Full and reduced logistic regression models were fit to identify significant predictors of postoperative hypoxemia, and test-set performance was assessed using misclassification error.

Key Findings

References

Key Takeaway: Male patients with high BMI and high glucose may benefit from closer postoperative monitoring for early detection and timely intervention of hypoxemia.

Key Risk Factors

- **Higher BMI** elevates odds of postoperative hypoxemia , aligning with established literature⁴
- **Higher glucose level** were significantly associated with increased odds of hypoxemia
- **Male** patients have higher odds of hypoxemia, which contradicts previous findings that showed being female as a risk factor⁴

Model Performance and Limitation

- The full model achieved a lower test misclassification rate (37%) than the reduced model (43%), suggesting that including a broader set of lab test results is necessary to capture the complexity of hypoxemia
- Only [percentage]% of patients reported postoperative hypoxemia, likely contributing to the relatively high misclassification rate

Future Directions

- Employing oversampling or re-weighting techniques to better account for imbalance and improve predictive accuracy
- Incorporating a wider variety of laboratory tests to strengthen predictive power
- Exploring how hypoxemia interacts with other concurrent postoperative complications

Introduction & Literature Review

- Postoperative hypoxemia is a frequent complication from anesthetic¹
- The MOVER EPIC dataset includes records from 64,354 surgeries between 2017 and 2022² and can be used to study this issue
- Prior research has assessed how postoperative hypoxemia is affected by different variables, but these models only contained up to four laboratory test results^{3,4,5}
- To fill this gap characterized by few laboratory test results, this work modelled postoperative hypoxemia probability using logistic regression with up to ten preoperative laboratory test results and three demographic variables as covariates
- The objective was to detect significant predictors of postoperative hypoxemia and determine the accuracy of predicting its probability with just these variables

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